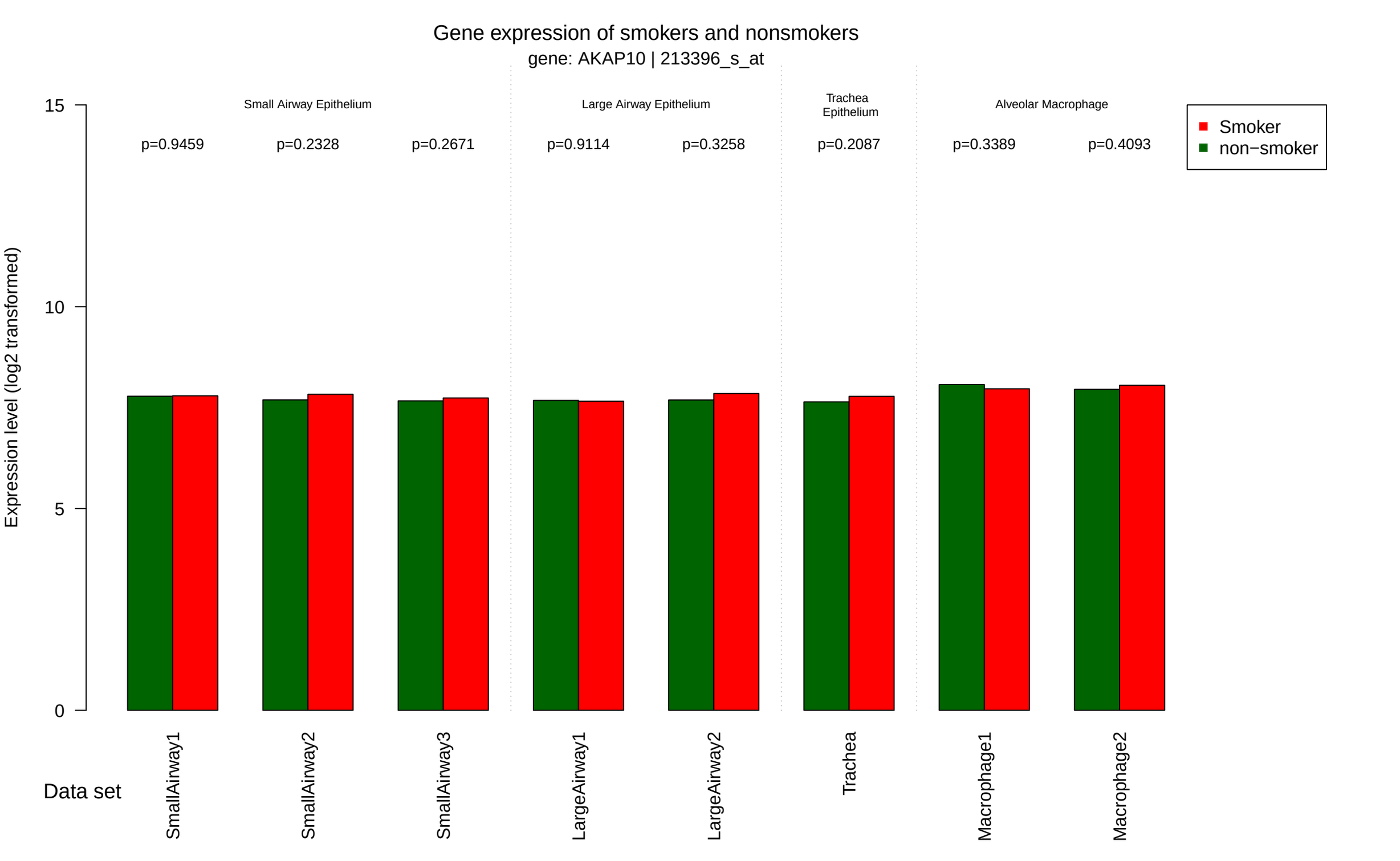
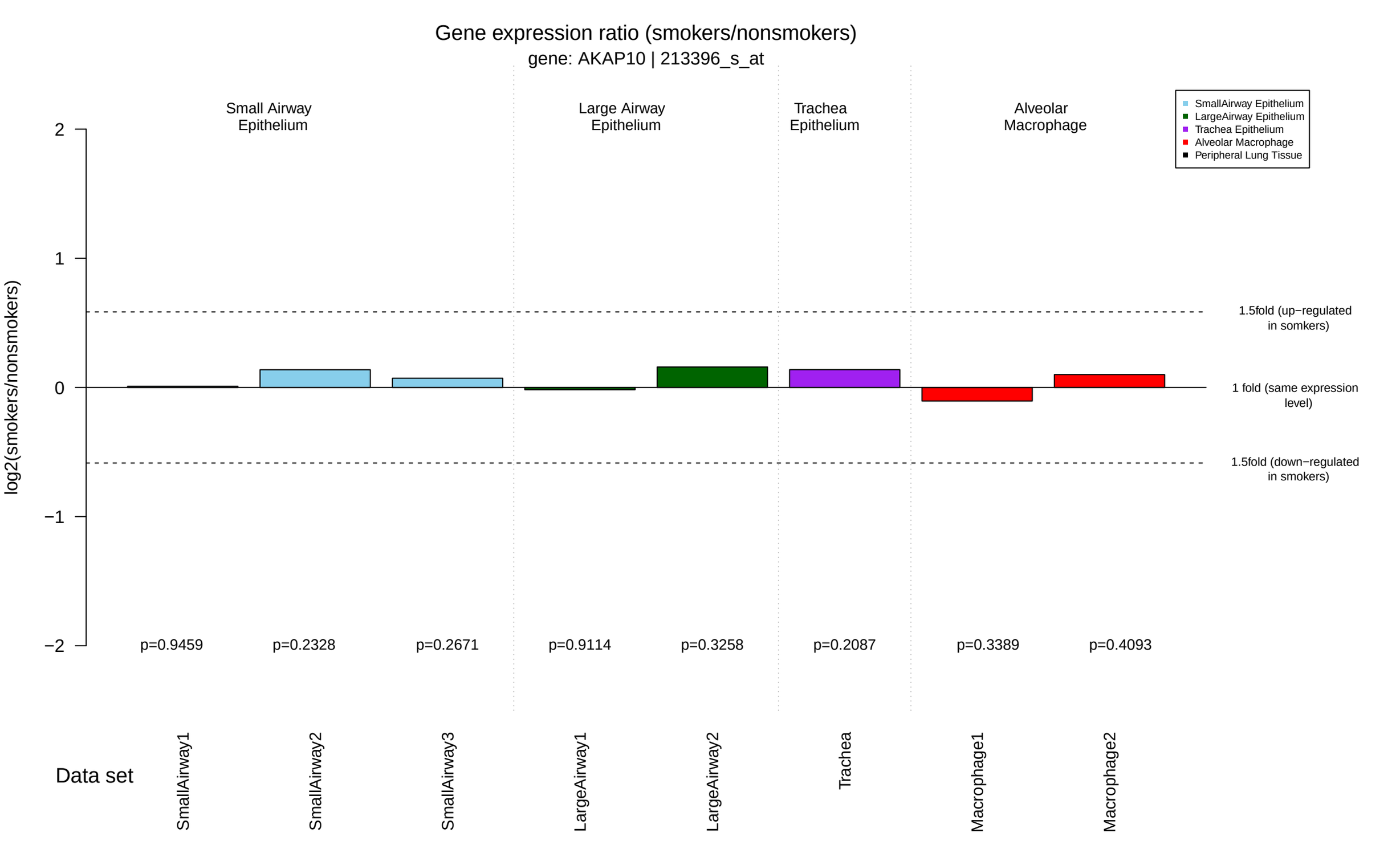
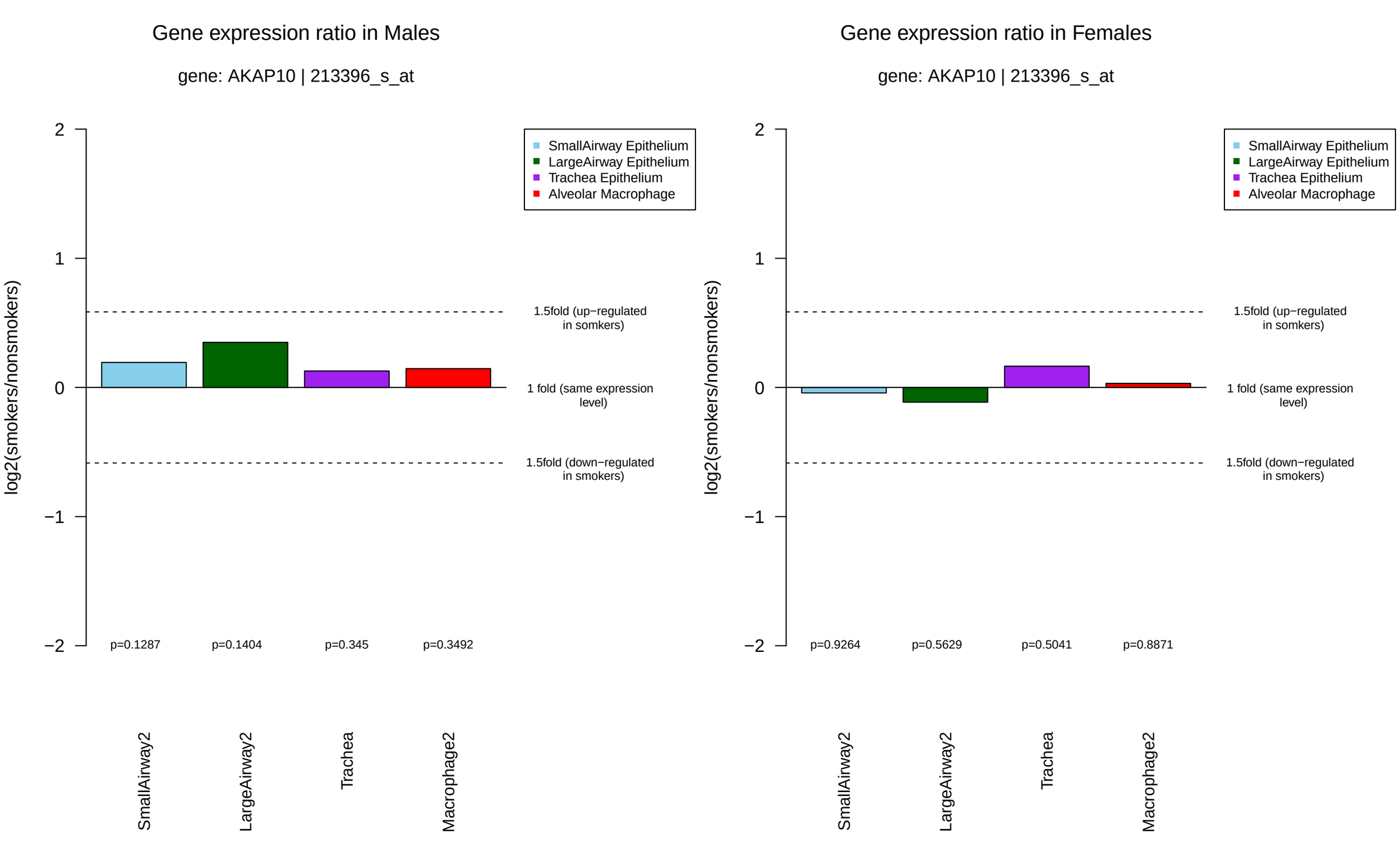
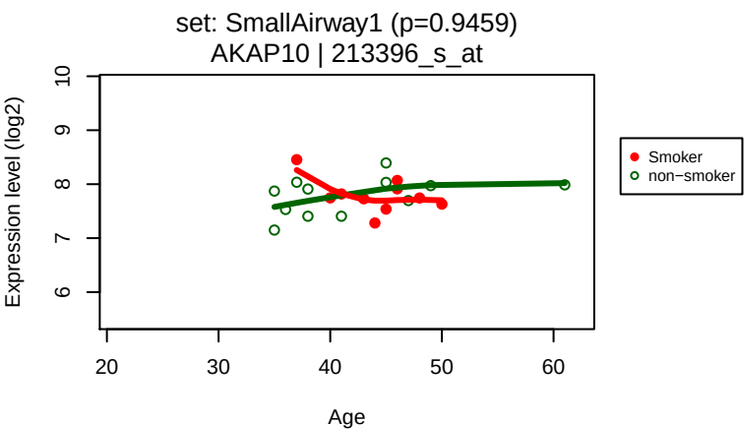
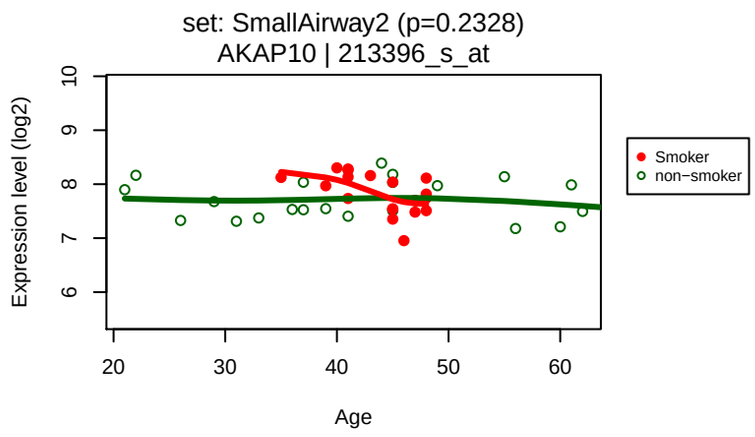
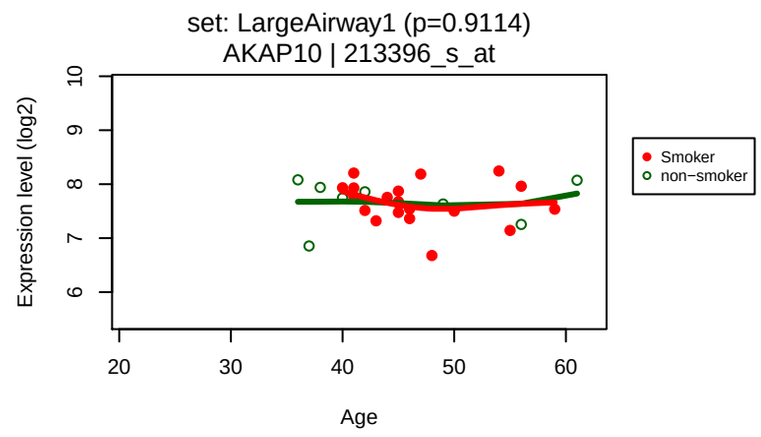

Gene expression of smokers and nonsmokers
gene: AKAP10 | 213396_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9459
p=0.2328
p=0.2671
p=0.9114
p=0.3258
p=0.2087
p=0.3389
p=0.4093
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

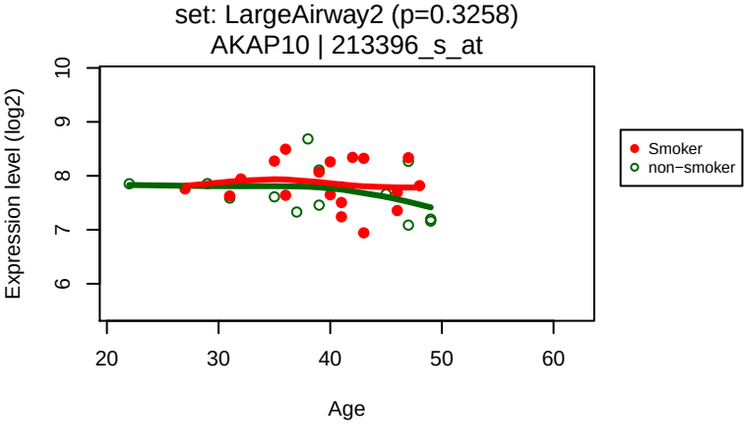

Gene expression ratio (smokers/nonsmokers)
gene: AKAP10 | 213396_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9459
p=0.2328
p=0.2671
p=0.9114
p=0.3258
p=0.2087
p=0.3389
p=0.4093
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

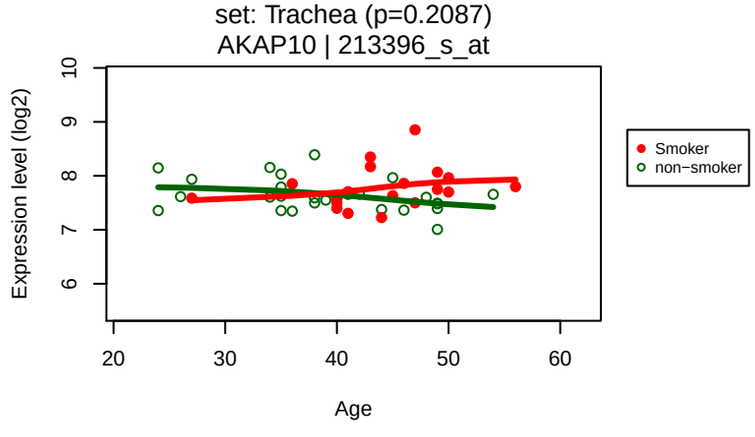

Gene expression ratio in Males
gene: AKAP10 | 213396_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1287
p=0.1404
p=0.345
p=0.3492
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP10 | 213396_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9264
p=0.5629
p=0.5041
p=0.8871
SmallAirway2
LargeAirway2
Trachea
Macrophage2

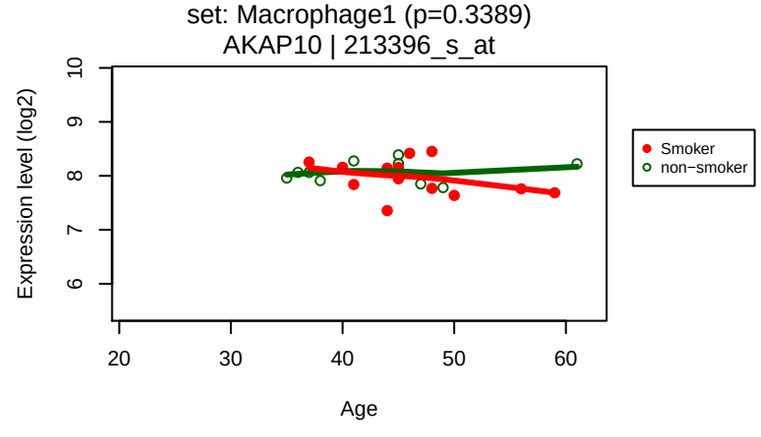

set: SmallAirway1 (p=0.9459)
AKAP10 | 213396_s_at
Expression level (log2)
Age
Smoker
non−smoker

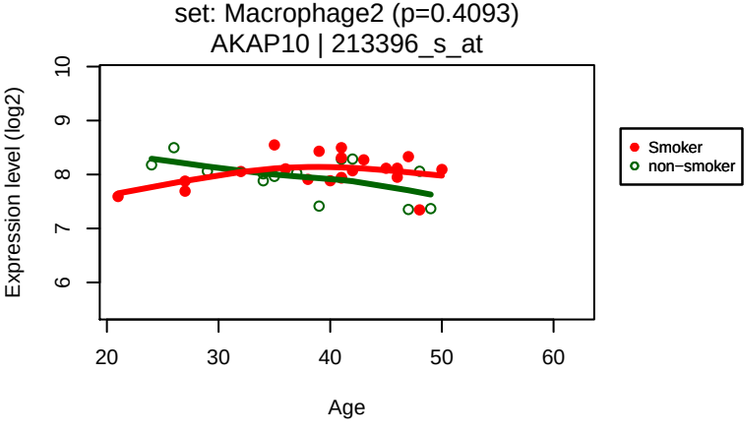

set: SmallAirway2 (p=0.2328)
AKAP10 | 213396_s_at
Expression level (log2)
Age
Smoker
non−smoker

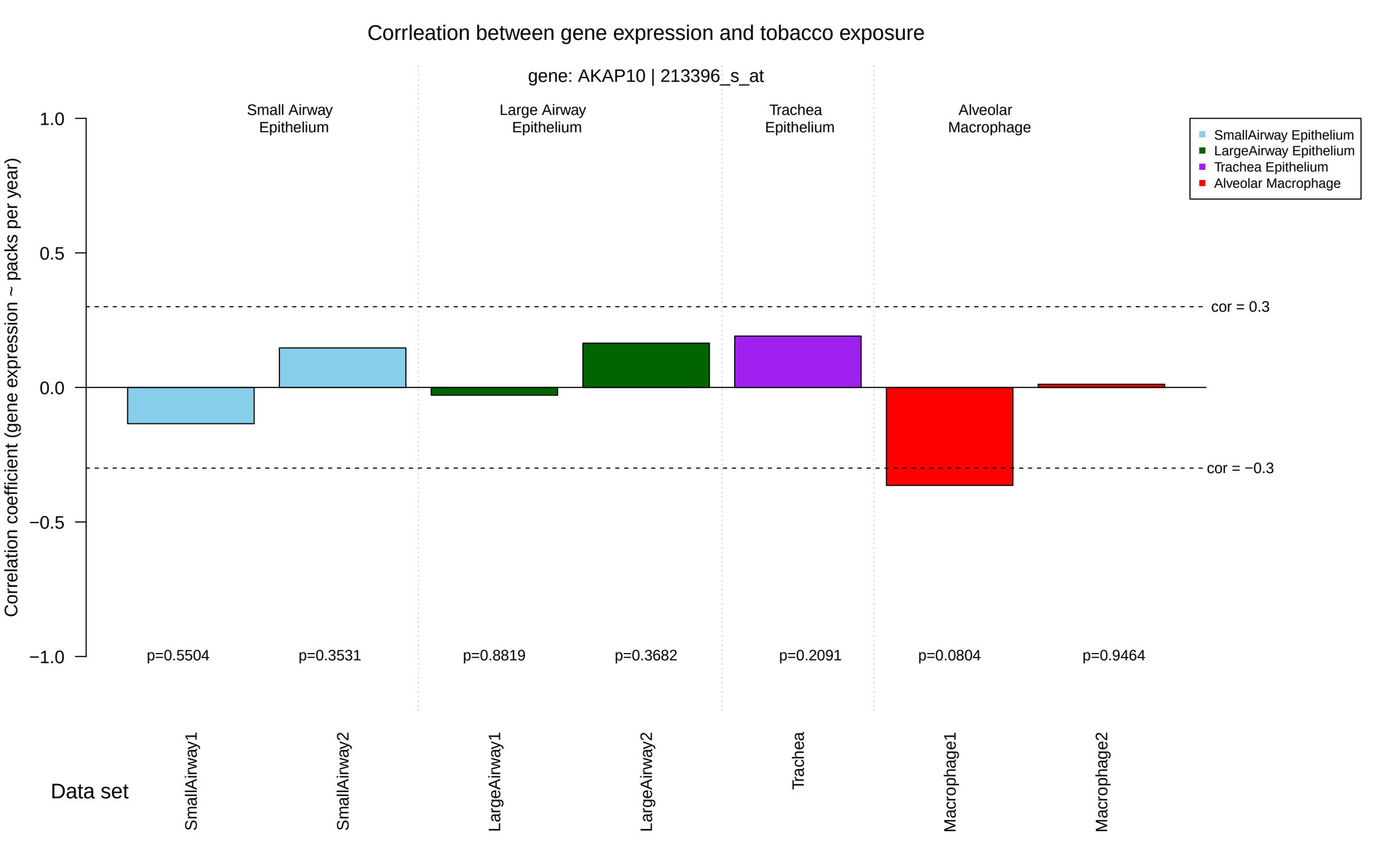

set: LargeAirway1 (p=0.9114)
AKAP10 | 213396_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3258)
AKAP10 | 213396_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2087)
AKAP10 | 213396_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3389)
AKAP10 | 213396_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4093)
AKAP10 | 213396_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP10 | 213396_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5504
p=0.3531
p=0.8819
p=0.3682
p=0.2091
p=0.0804
p=0.9464
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2